

DAVID SWAN

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SUMMARY

Software engineer with 4 years at Lockheed Martin Space building autonomy and edge-inference systems for space hardware, currently leading C++ and Python flight software and on-board ML model serving on FPGAs. Also built ROS2 navigation for autonomous rovers and co-founded NexusOdds, a SaaS with 1,290+ lifetime paying customers.

WORK EXPERIENCE

Lockheed Martin Space

Palo Alto, CA

AI Research Lead Engineer

Mar 2026 – Present

- Lead development of **C++** and **Python** flight software that runs fiber-optic loopback bit error rate (BER) tests at 25.875 Gbps and sends the results to the ground as telemetry
- Own the **Yocto Project** builds of the custom embedded **Linux** images that run on flight hardware
- Drive on-board model serving with **Ray Serve**, quantizing models and running multiple models concurrently on **FPGA** boards
- Architected and deployed a complete autonomous rover navigation system in **ROS2**, integrating LIO-SAM for 3D SLAM and a tuned Nav2 stack for motion control
- Delivered real-time open-vocabulary object detection and text-based waypoint navigation (custom **ROS2** Python package) on a tight deadline, helping win new business

AI Research Engineer

Mar 2024 – Mar 2026

- Achieved a 1500× inference speedup for an automatic target recognition (ATR) model on low-power (low-SWaP), path-to-flight hardware by quantizing and compiling it, then recovered full accuracy with quantization-aware training
- Accelerated super-resolution and object detection inference 15× on AMD and NVIDIA platforms with INT8 quantization, pruning and hardware-specific optimization in **Vitis AI** and **CUDA**
- Led development of applications that interface with proprietary flight software, using LiDAR and electro-optical data with deep learning models for real-time target identification and tracking
- Built real-time plotting tools in **Python** using multiprocessing and sockets to visualize live data streams
- Established ground-test infrastructure for ML algorithms across a suite of single-board computers to de-risk satellite deployments

Associate AI Research Engineer

Aug 2022 – Mar 2024

- Contributed to a **Gazebo** space simulation for collaborative motion planning on a dual UR10e robotic arm testbench (**ROS1**)
- Built a satellite orchestration platform web app with **React**, **JavaScript** and **Tailwind CSS**

Intern

Jan 2022 – May 2022

- Fine-tuned an object detection model in **PyTorch** and deployed it on an NVIDIA Jetson, reaching 90% mAP detecting people from drone imagery

NexusOdds

nexusodds.com

Co-Founder

May 2024 – Present

- Launched a sports betting analytics SaaS that has grown to 2,600+ registered users and 1,290+ lifetime paying customers, covering 33+ sportsbooks and prediction markets
- Designed the original **Python** data-collection framework and serverless **AWS** ingestion pipeline; the platform now processes ~32M odds records a day (~1B a month, 435K in a peak minute)

EDUCATION

Georgia Institute of Technology

Dec 2025

Master of Science in Computer Science, Specialization in Interactive Intelligence; completed while working full time

The Pennsylvania State University

May 2022

Bachelor of Science in Computer Engineering, Minor in Mathematics

SKILLS

Languages: C++ (embedded), Python, TypeScript, JavaScript

Robotics and Autonomy: ROS2, ROS1, Nav2, 3D SLAM (LIO-SAM), Gazebo, LiDAR/EO perception, object tracking

ML Inference: PyTorch, TensorFlow, TensorRT, CUDA, Vitis AI, Ray Serve, INT8 quantization, QAT, pruning

Embedded and Hardware: Embedded Linux, Yocto, FPGA (AMD Xilinx Versal, UltraScale+), NVIDIA Jetson Orin/Xavier

Tools and Cloud: Linux, Git, Docker, AWS, React, Node.js